Air University

Final Semester Examinations: Spring 2025

Student ID: [redacted]

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(To be solved on Answer Books only)

Subject: APPLIED PHYSICS
Class: BSCYS-II (EVENING)
Sections: A & B
Course Code: Ph-111 /102
Paper: A

Time Allowed: 3 Hrs.
Max Marks: 100
FM's Name: NAZIR AHMED MALIK
FM's Signature:

INSTRUCTIONS

- Attempt responses on the answer book only.
- Nothing is to be written on the question paper.
- Rough work or writing on question paper will be considered as use of unfair means.
- Calculators are allowed.

Q1. (CLO-1/ PLO-2/C-2) **(Marks: 10x3 = 30)**

a) State: i) Ampere's law ii) luminous and non-luminous objects

iii) Faraday's law of electromagnetic induction iv) Dispersion

b) Define resistivity. Write the factors affecting resistance in a conductor.

c) Define polarization? Describe the polarization of light by selective absorption?

Q2. (CLO-2/ PLO-2/C-2) **(Marks: 40)**

a) While understanding the Bio-Savart law, derive the formula of magnetic field due to a **long straight wire** carrying current at a distance "**d**" from it. **(Marks: 15)**

b) While understanding the Ampere's law, derive the formula of magnetic field inside a **long solenoid** carrying current '**I**' at its centre. **(Marks: 15)**

c) State Faraday-Maxwell Law and derive its relation. **(Marks: 10)**

Q3. (CLO-3/ PLO-2/C-3) **(Marks: 10x3 = 30)**

a) A small but measureable current of 1.2×10^{-10} A exists in a copper wire whose diameter is 3.0 mm. The number of charge carriers per unit volume is 8.49×10^{28} m^{-3}. Assuming the current is uniform calculate the current density and electron drift velocity.

b) A uniform magnetic field **B**, with magnitude 1.2 mT, is directed vertically upward throughout the volume of a chamber in which a proton enters in horizontal direction from south to north with kinetic energy of 5.3 MeV. What force acts on the proton when it enters in the chamber also find its acceleration?

c) A screen is separated from the double slit source by 1.2 m. The distance between the two slits is 0.030mm. A second order bright fringe is located at 4.5 cm from the center line. Find the width of the bright fringe.

BEST OF LUCK